A Cache Shared by Many Servers

Distributed caching and stale data

SCAN / LEARN

THE MAIN IDEA

A cache keeps reusable results close to where they are needed. A shared cache can help a server fleet, but old or incorrect entries can spread mistakes.

FOLLOW THE SYSTEM

01 ASK

An application requests a stored result.

02 LOOKUP

The cache checks whether it has a valid copy.

03 HIT / MISS

Reuse the entry, or ask the original source.

04 REFRESH

A policy expires or updates old entries.

REQUEST → CACHE → REUSE OR FETCH → REFRESH

COMMON MISCONCEPTION

Caching every answer is always safe and cheaper.

MORE ACCURATE

Freshness, privacy and storage costs determine safe reuse.

THREE QUESTIONS TO KEEP

PROBLEM: How can repeated reads avoid repeated work?

FAILURE: An old cached price misleads a customer.

IN REAL LIFE: Web apps, databases and AI workflows.

SOLO DISCOVERY

Write 3 changing facts. Which are safe to reuse tomorrow?